

Problems for week 10

Solutions

Problem 1. Let C be the unit circle in the plane, oriented counterclockwise, and consider the vector field $\mathbf{F} = x^2\hat{\mathbf{i}} + xy\hat{\mathbf{j}}$. Find the flux of $\mathbf{F}$ through C by directly evaluating the line integral $\oint_C \mathbf{F} \cdot \hat{\mathbf{n}} \, ds$.

Solution 1. The normal vector to C at (x, y) is $\hat{\mathbf{n}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$. Therefore $\mathbf{F} \cdot \hat{\mathbf{n}} = (x^2, xy) \cdot (x, y) = x^3 + xy^2 = x(x^2 + y^2) = x$ (recall that $x^2 + y^2 = 1$ on the unit circle).

We parametrise the unit circle by the polar angle θ : so $x = \cos \theta$, $y = \sin \theta$, and the length element is $ds = d\theta$. Therefore

$$\oint_C \mathbf{F} \cdot \hat{\mathbf{n}} \, ds = \oint_C x \, ds = \int_0^{2\pi} \cos \theta \, d\theta = 0.$$

Problem 2. Find a parametric representation for the following surfaces:

- (a) The part of the sphere $x^2 + y^2 + z^2 = 4$ that lies above the cone $z = \sqrt{x^2 + y^2}$;
- (b) The part of the cylinder $x^2 + z^2 = 9$ that lies above the xy -plane and between the planes $y = -4$ and $y = 4$.

Solution 2.

- (a) The cone intersects the sphere in the circle $x^2 + y^2 = 2, z = \sqrt{2}$, and we want the portion of the sphere above this. Using x, y as parameters, we can parametrise the surface as $x = x, y = y, z = \sqrt{4 - x^2 - y^2}$, where $x^2 + y^2 \leq 2$.
Alternatively, in cylindrical coordinates, using r, θ as parameters, we can parametrise the surface by $x = r \cos \theta, y = r \sin \theta, z = \sqrt{4 - r^2}$, where $0 \leq r \leq \sqrt{2}, 0 \leq \theta \leq 2\pi$.
Or, using spherical coordinates, where the sphere is $\rho = 2$ and the cone is $\phi = \frac{\pi}{4}$: using ϕ, θ as parameters, the parametrisation becomes $x = 2 \sin \phi \cos \theta, y = 2 \sin \phi \sin \theta, z = 2 \cos \phi$, where $0 \leq \phi \leq \frac{\pi}{4}$ and $0 \leq \theta \leq 2\pi$.
- (b) $x^2 + z^2 = 9$ is the cylinder of radius 3 centred on the y -axis; its projection to the xz -plane is the circle of radius 3, easiest to parametrise by $x = 3 \cos \theta, z = 3 \sin \theta$. The half where $z \geq 0$ (above the xy -plane) corresponds to $0 \leq \theta \leq \pi$. Meanwhile, y varies between -4 and 4 . Hence: $x = 3 \cos \theta, y = y, z = 3 \sin \theta, -4 \leq y \leq 4, 0 \leq \theta \leq \pi$.
Or, using rectangular coordinates, since we are only considering the upper half of the cylinder where $z = \sqrt{9 - x^2}$: $x = x, y = y, z = \sqrt{9 - x^2}, -3 \leq x \leq 3, -4 \leq y \leq 4$.

Problem 3. Find the area of the surface S , where S is the part of the surface $z = 4 - 2x^2 + y$ that lies above the triangle with vertices $(0, 0, 0)$, $(1, 0, 0)$ and $(1, 1, 0)$.

Solution 3. $z = f(x, y) = 4 - 2x^2 + y$ over the triangle D with vertices $(0, 0, 0)$, $(1, 0, 0)$ and $(1, 1, 0)$ (i.e. $0 \leq y \leq x, 0 \leq x \leq 1$ in the xy -plane). Its area is given by

$$\text{Area}(S) = \iint_D \sqrt{1 + (f_x)^2 + (f_y)^2} \, dA = \iint_D \sqrt{1 + (-4x)^2 + 1^2} \, dA = \int_0^1 \int_0^x \sqrt{2 + 16x^2} \, dy \, dx.$$

Inner integral: $\left[y\sqrt{2 + 16x^2} \right]_0^x = x\sqrt{2 + 16x^2}.$

Outer integral: $\int_0^1 x\sqrt{2 + 16x^2} \, dx = \left[\frac{1}{48}(2 + 16x^2)^{3/2} \right]_0^1 = \frac{1}{48}(18^{3/2} - 2^{3/2}) = \frac{1}{48}(54\sqrt{2} - 2\sqrt{2}) = \frac{13\sqrt{2}}{12}.$

Problem 4. Evaluate the following surface integrals:

(a)

$$\iint_S y^2 \, dS,$$

where S is the part of the sphere $x^2 + y^2 + z^2 = 1$ that lies above the cone $z = \sqrt{x^2 + y^2}$.

(b)

$$\iint_S (x + y + z) \, dS,$$

where S is the part of the half-cylinder $x^2 + z^2 = 1, z \geq 0$, that lies between the planes $y = 0$ and $y = 2$.

Solution 4.

(a) In spherical coordinates, we have $dS = \sin \phi \, d\phi \, d\theta$; and $y^2 = (\sin \phi \sin \theta)^2$. The portion of sphere we are considering corresponds to $0 \leq \phi \leq \pi/4$. Hence

$$\iint_S y^2 \, dS = \int_0^{2\pi} \int_0^{\pi/4} (\sin \phi \sin \theta)^2 \sin \phi \, d\phi \, d\theta = \left(\int_0^{\pi/4} \sin^3 \phi \, d\phi \right) \left(\int_0^{2\pi} \sin^2 \theta \, d\theta \right).$$

We have that

$$\int_0^{\pi/4} \sin^3 \phi \, d\phi = \int_0^{\pi/4} (1 - \cos^2 \phi) \sin \phi \, d\phi = \left[-\cos \phi + \frac{1}{3} \cos^3 \phi \right]_0^{\pi/4} = \frac{2}{3} - \frac{5\sqrt{2}}{12}, \text{ and}$$

$$\int_0^{2\pi} \sin^2 \theta \, d\theta = \int_0^{2\pi} \frac{1}{2}(1 - \cos 2\theta) \, d\theta = \pi, \text{ so } \iint_S y^2 \, dS = \left(\frac{2}{3} - \frac{5\sqrt{2}}{12} \right) \pi.$$

- (b) Parametrise the cylinder by $x = \cos t, y = y, z = \sin t$: then $dS = dy dt$ by calculating $|\mathbf{r}_y \times \mathbf{r}_t| = |(\cos t, 0, \sin t)| = 1$. The surface S corresponds to $0 \leq y \leq 2$ and $0 \leq t \leq \pi$. Then

$$\iint_S (x+y+z) dS = \int_0^2 \int_0^\pi (\cos t + y + \sin t) dt dy = \int_0^2 (\pi y + 2) dy = \left[\frac{1}{2} \pi y^2 + 2y \right]_0^2 = 2\pi + 4.$$

Problem 5. Evaluate the surface integral

$$\iint_S \mathbf{F} \cdot d\mathbf{F}$$

for the given vector field $\mathbf{F}$ and oriented surface S . In other words, find the flux of $\mathbf{F}$ through S .

- (a) $\mathbf{F}(x, y, z) = xy\hat{i} + yz\hat{j} + zx\hat{k}$, S is the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the square $0 \leq x \leq 1, 0 \leq y \leq 1$, and has upward orientation.
 (b) $\mathbf{F}(x, y, z) = -x\hat{i} - y\hat{j} + z^3\hat{k}$, S is the part of the cone $z = \sqrt{x^2 + y^2}$ between $z = 1$ and $z = 3$ with downward orientation.
 (c) $\mathbf{F}(x, y, z) = y\hat{i} - x\hat{j} + 2z\hat{k}$, S is the hemisphere $x^2 + y^2 + z^2 = 4, z \geq 0$, oriented downward.

Solution 5.

- (a) $\mathbf{F}(x, y, z) = (xy, yz, zx)$, and since $z = g(x, y) = 4 - x^2 - y^2$, for the upward orientation we have

$$d\mathbf{S} = (-g_x, -g_y, 1) dx dy = (2x, 2y, 1) dx dy.$$

Hence

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_S (2x^2y + 2y^2z + xz) dx dy \\ &= \int_0^1 \int_0^1 (2x^2y + 2y^2(4 - x^2 - y^2) + x(4 - x^2 - y^2)) dx dy. \end{aligned}$$

Inner integral: $\int_0^1 -2y^4 + (8 - 2x^2 - x)y^2 + 2x^2y + (4x - x^3) dy = -\frac{2}{5} + \frac{1}{3}(8 - 2x^2 - x) + x^2 + (4x - x^3).$

Outer integral: $\int_0^1 (-x^3 + \frac{1}{3}x^2 + \frac{11}{3}x + \frac{34}{15}) dx = -\frac{1}{4} + \frac{1}{9} + \frac{11}{6} + \frac{34}{15} = \frac{713}{180}.$

- (b) $\mathbf{F}(x, y, z) = (-x, -y, z^3)$, and since $z = g(x, y) = \sqrt{x^2 + y^2}$, with the downward orientation we have

$$d\mathbf{S} = -(-g_x, -g_y, 1) dx dy = \left(\frac{x}{\sqrt{x^2 + y^2}}, \frac{y}{\sqrt{x^2 + y^2}}, -1 \right) dx dy.$$

Moreover, the surface S lies above the annulus $1 \leq \sqrt{x^2 + y^2} \leq 3$ in the xy -plane (since the cone intersects $z = 1$ and $z = 3$ along circles of radii 1 and 3). Hence

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \left(\frac{-x^2 - y^2}{\sqrt{x^2 + y^2}} - z^3 \right) dx dy = \iint_S (-r - z^3) r dr d\theta = \int_0^{2\pi} \int_1^3 (-r - r^3) r dr d\theta$$

(using $z = r$ everywhere on S).

$$\text{Inner integral: } \int_1^3 (-r - r^3) r dr = \left[-\frac{1}{3}r^3 - \frac{1}{5}r^5 \right]_1^3 = -\frac{1}{3}(27 - 1) - \frac{1}{5}(243 - 1) = -\frac{856}{15}.$$

$$\text{Outer integral: } 2\pi \cdot \frac{-856}{15} = -\frac{1712\pi}{15}.$$

(c) The normal vector points radially inwards, i.e. it is negatively proportional to (x, y, z) . Since $|(x, y, z)| = \sqrt{x^2 + y^2 + z^2} = 2$ on S , we have $\hat{\mathbf{n}} = -\frac{1}{2}(x, y, z)$. Hence

$$\mathbf{F} \cdot \hat{\mathbf{n}} = (y, -x, 2z) \cdot \left(-\frac{1}{2}(x, y, z)\right) = -z^2.$$

We parametrise the spherical surface using the angles ϕ and θ (upper half: $0 \leq \phi \leq \frac{\pi}{2}$, $0 \leq \theta \leq 2\pi$), so that $-z^2 = -(2 \cos \phi)^2$ and $dS = 2^2 \sin \phi d\phi d\theta$. Therefore

$$\begin{aligned} \iint_S \mathbf{F} \cdot d\mathbf{S} &= \iint_S -z^2 dS = \int_0^{2\pi} \int_0^{\pi/2} -16 \cos^2 \phi \sin \phi d\phi d\theta \\ &= -32\pi \int_0^{\pi/2} \cos^2 \phi \sin \phi d\phi = -32\pi \left[-\frac{1}{3} \cos^3 \phi \right]_0^{\pi/2} = -\frac{32\pi}{3}. \end{aligned}$$